

UNIVERSITY OF THE WITWATERSRAND
SCHOOL OF BUSINESS SCIENCES

INFORMATION SYSTEMS IIA (INFO2001A)
TEST 1 – AUGUST 2025

DURATION: 1¾ hours (105 minutes)

MARKS: 55

INSTRUCTIONS:

1. This paper consists of **FIVE (5)** pages. Please ensure that you have them all.
2. You must answer all questions and number them correctly.
3. Sections A and B must be answered in the "**YELLOW**" booklet, while Section C must be answered in the "**GREEN**" booklet. Also put the Section(s) you are answering in the space for "**PAPER No.:**" on each booklet.
4. You may draw diagrams in pencil.
5. You may draw diagrams using landscape page orientation if necessary.
6. You may **NOT** write in the right-hand margins – These are for marking purposes only.

Section A (10 marks, 15 minutes)

Each of the following questions consists of a statement followed by four options (A, B, C, D). Select the **ONE (1)** option that best answers the question.

1. Which of the following Object-Oriented Associations is supported by relational databases?
A. Aggregation
B. Composition
C. Inheritance
D. None of the above
[1]
2. Designing a database involves multiple steps. Which of the following shows the correct sequence?
A. Conceptual, Logical, Physical
B. Physical, Conceptual, Logical
C. Conceptual, Physical, Logical
D. Any of the above sequence is correct
[1]
3. In Relational Database Management Systems, different classes of relations are created using the _____ technique to prevent modification anomalies.
A. Functional dependencies
B. Data integrity
C. Referential integrity
D. Normal forms
[1]
4. Data which improves the performance and accessibility of the database are called _____.
A. Indexes
B. User data
C. Application metadata
D. Data dictionary
[1]

5. Consider the following dependencies and the BOOK table in a relational database design. Determine the current normal form of the given relation.

ISBN $\Rightarrow$ Title, ISBN $\Rightarrow$ Publisher, Publisher $\Rightarrow$ Address

- A. Unnormalised Form
- B. First Normal Form
- C. Second Normal Form
- D. Third Normal Form

[1]

6. In Relational Database Management Systems, the constraint that no key attribute (column) may contain a NULL value is referred to as _____.

- A. Referential integrity
- B. Multi-valued dependency
- C. Entity integrity
- D. Functional dependency

[1]

7. A table has fields InvoiceNumber, ProductName, SaleDate and ProductPrice, where the domains of these fields include only atomic values, only the following functional dependencies and those that can be inferred from them:

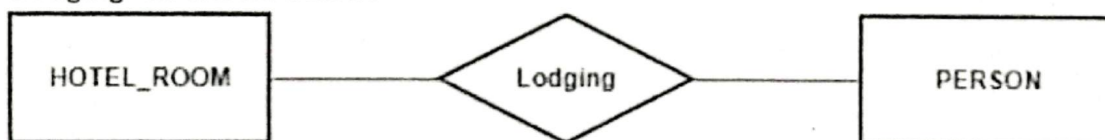
InvoiceNumber $\Rightarrow$ SaleDate, ProductName $\Rightarrow$ ProductPrice

In terms of normalisation, this table is in _____.

- A. Unnormalised form
- B. First normal form, but not in second normal form
- C. Second normal form, but not in third normal form
- D. Third normal form

[1]

8. Consider the entities "HOTEL_ROOM", and "PERSON" with a many-to-many relationship "Lodging" as shown below:



If you wish to store data about the rent payment to be made by person(s) occupying different hotel rooms, then this data should appear as an attribute of _____.

- A. PERSON
- B. HOTEL_ROOM
- C. Lodging
- D. None of the above

[1]

9. At a South African university, each student is assigned a *student number* by the registrar's office (e.g., 2025123456). The database also generates a unique *StudentID* as an auto-increment value for internal use.

Which statement best describes the keys in this situation?

- A. Both the student number and the StudentID are natural keys.
- B. Both the student number and the StudentID are surrogate keys.
- C. The student number is a surrogate key, and the StudentID is a natural key.
- D. The student number is a natural key, and the StudentID is a surrogate key.

[1]

10. A university database stores student enrolments in a single table as follows:

StudentID	StudentName	CourseCode	CourseName	Lecturer
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Consider the following situations:

1. Removing the last student enrolled in a course also deletes data about the course itself.
2. Changing the lecturer's name for a course requires updating multiple rows for every student in that course.
3. You cannot add a new course to the table unless at least one student is enrolled in it.

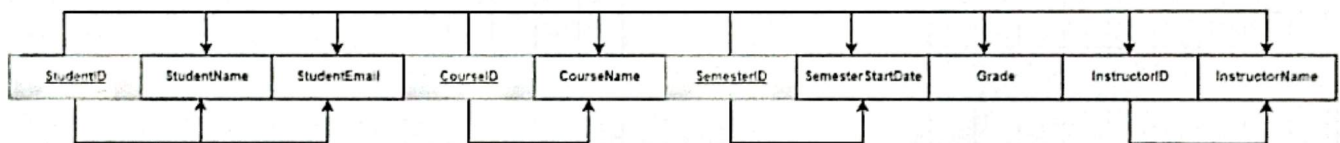
Which types of anomalies do these situations represent?

- A. 1 → Deletion anomaly, 2 → Update anomaly, 3 → Insertion anomaly
- B. 1 → Insertion anomaly, 2 → Deletion anomaly, 3 → Update anomaly
- C. 1 → Update anomaly, 2 → Insertion anomaly, 3 → Deletion anomaly
- D. 1 → Update anomaly, 2 → Deletion anomaly, 3 → Insertion anomaly

[1]

Section B (25 marks, 50 minutes)

Consider the following dependency diagram for an unnormalised database structure being used by a university to store data about its enrolments: **Note:** Assume that all fields contain atomic values.



1. List and classify each dependency shown in the dependency diagram. [5]
2. Normalise the database structure to second normal form (2NF) using fully labelled dependency diagrams. [7]
3. Extend your answer for Question 1 and normalise the structure to third normal form (3NF) using fully labelled dependency diagrams. [3]

Note: You do NOT have to redraw any unchanged dependency diagrams from Question 2.

4. Use your answer for Question 3 and the business rules stated below to draw a fully labelled entity relationship diagram (ERD) for the normalised database structure.

A student must be enrolled in at least one (1) course in order to be considered a student. A course may have many students enrolled, but each course must have at least one (1) student. A course must be offered in at least one (1) semester. A semester may include many courses, but a semester may also exist without any courses being offered. An enrolment record must always link exactly one student, one course, and one semester. Each student-course-semester combination must appear in ENROLMENT if a student takes that course in that semester. Each enrolment must be supervised by one (1) instructor. An instructor may supervise many enrolments, but an instructor may also exist without supervising any enrolment. Students are identified using a unique StudentID. Courses are identified using a

unique CourseID. Semesters are identified using a unique SemesterID. Instructors are identified using a unique InstructorID.

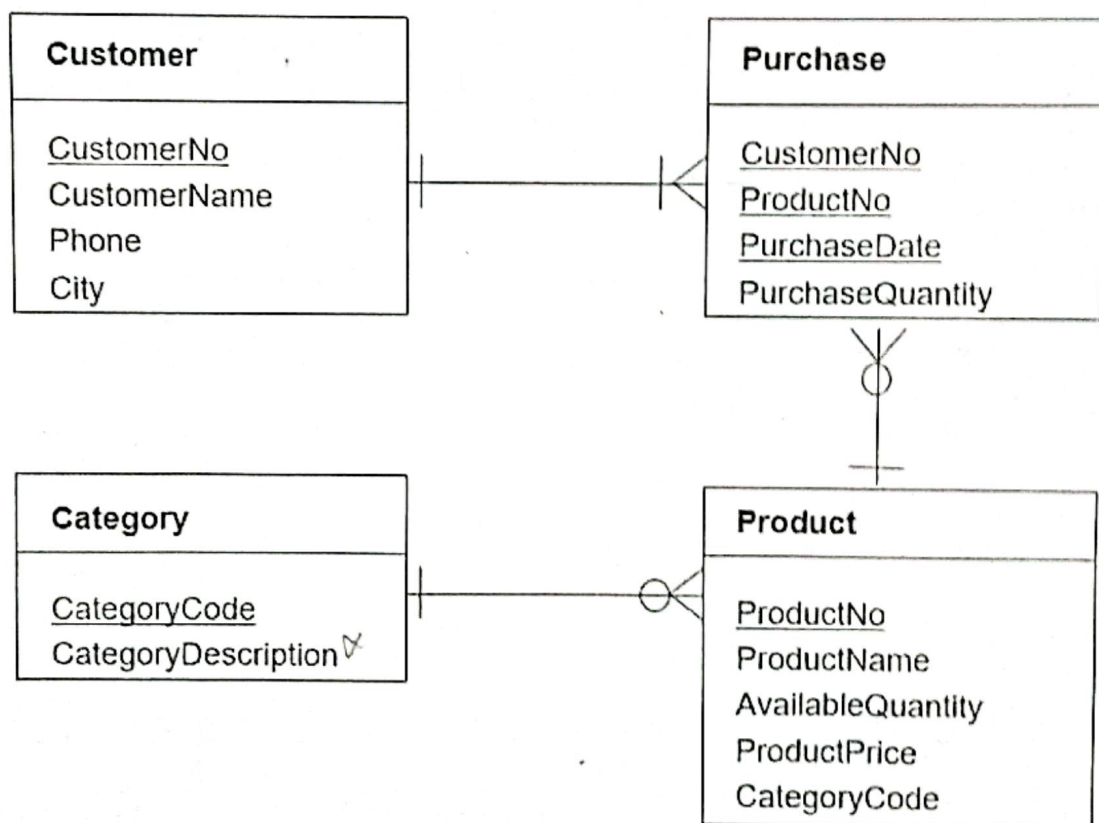
Include all attributes and label all primary and foreign keys.

[10]

Section C (20 marks, 40 minutes)

Consider the following ERD diagram. It represents a database that a small retail company uses to track the purchases that have been made by customers. Each customer may have made more than one (1) purchase, but a particular purchase would have been made by only one (1) customer. Each purchase is only for one (1) product, but a product can be purchased more than once. There are different product categories, and each product falls into one (1) category, whereas there can be many products in a category.

Please note that Questions 1 and 2 are based on the following ERD diagram, so carefully study it before attempting these questions.



The datatypes of the fields in this database are as follows:

Character: CustomerName, Phone, City, ProductName, CategoryCode, CategoryDescription

Number: CustomerNo, ProductNo, PurchaseQuantity, AvailableQuantity, ProductPrice

Date: PurchaseDate

1. Write an SQL query to find the total value of inventory on hand for each product category. The value for a single product is calculated as its available quantity multiplied by its price. The report should show the category code, the full category description, and the total inventory value for that category (heading of this column should be "Total Value"). Show only categories with a total inventory value greater than 10 000. [8]
2. The marketing team is preparing a targeted campaign for loyal customers in Cape Town. Write an SQL statement that provides a list of all products recently purchased by these customers. The report should show the product name, customer name, and purchase date. The date should be printed out in the same format as "3-June-2005" (heading of this column should be "Purchase Date"). Sort the results by the purchase date, showing the most recent purchases first. [8]
3. Which SQL keyword is used to remove duplicate rows in the result set?
 - A. DISTINCT
 - B. UNIQUE
 - C. DELETE
 - D. All of the above[1]
4. If one or more columns are present in a SELECT statement in addition to the SUM or AVG function, these columns need to be a part of _____. [1]
 - A. HAVING clause
 - B. GROUP By clause
 - C. ORDER By clause
 - D. WHERE clause
5. Explain the difference between NULL and 0 in SQL. [2]

Total Marks = 55

END OF TEST
